MHF4U Advanced Functions - Unit 1 Practice Test, Version 4

Worked key and thinking guide

60 marks: 10 multiple choice + 50 multipart. Award equivalent correct reasoning. For sketches, check labelled axes, key points, and open or closed endpoints.

Part A - Multiple choice [10]

Question	Answer	Reason
1	C	The input 2 has two outputs.
2	B	A square-root input requires `7-x>=0`.
3	C	Include -2; exclude 5.
4	D	The vertex is `(-3,-2)`.
5	B	Replacing x by -x leaves `x^4+2` unchanged.
6	B	The factor 2 compresses x-coordinates by 1/2; the graph moves right 1.
7	C	Solve `y=3x+4` for x.
8	C	Use the branch for `x>=2`: `5-2=3`.
9	C	`(f(3)-f(1))/(3-1)=(9-1)/2=4`.
10	D	A tangent gives the instantaneous slope.

Part B - Multipart questions [50]

11. Piecewise functions [9]

a. Draw `y=x+4` from filled `(-4,0)` towards open `(0,4)`. Draw `y=4-2x` from filled `(0,4)` to filled `(2,0)`. Draw `y=x-4` from open `(2,-2)` to filled `(5,1)`. The first branch's open point is covered by the second branch's filled point. [4]

b. Domain `[-4,5]`; range `(-2,4]`. [2]

c. Increasing on `[-4,0]` and `(2,5]` (accept equivalent description of the rising pieces). [1]

d. No. For example, `T(-3)=T(1.5)=1`. One horizontal line crosses the graph at two inputs, so the inverse gives two outputs for input 1. [2]

Thinking move: Include or exclude an endpoint according to the dot on the graph, and test horizontal lines for an inverse.

12. Absolute value graph [6]

The vertex is `(1,6)` and the V opens downward. `-2|x-1|+6=0` gives `|x-1|=3`, so x-intercepts are `(-2,0)` and `(4,0)`. Domain `(-infinity,infinity)`; range `(-infinity,6]`. The negative sign reflects the parent graph in the x-axis; 2 stretches it vertically by factor 2. [3 graph; 2 domain/range; 1 explanation]

Thinking move: Solve for both absolute-value branches to find both intercepts.

13. Transforming a function [5]

a. Match the original input 16: `-4(x-3)=16`, so `x=-1`. The new output is `2f(16)+1=2(4)+1=9`; point `(-1,9)`. [2]

b. Require `-4(x-3)>=0`, giving domain `(-infinity,3]`. Since the square root is nonnegative, range `[1,infinity)`. [2]

c. Reflect horizontally in the y-axis, compress horizontally by factor 1/4, and shift right 3. Equivalently, the point map is `(u,v)->(3-u/4,2v+1)`. [1]

Thinking move: Set the entire new input equal to the old x-coordinate.

14. Inverse relations [5]

a. The original parabola has range `[3,infinity)`, so its inverse relation has domain `[3,infinity)` and range `(-infinity,infinity)`. [2]

b. The parabola fails the horizontal-line test: `q(-2)=4=q(0)`, so its inverse would give input 4 two outputs. [1]

c. Restrict the original domain to `x>=-1`. Then `q^(-1)(x)=-1+sqrt(x-3)` for `x>=3`. Also accept restriction `x<=-1` with `q^(-1)(x)=-1-sqrt(x-3)`. [2]

Thinking move: Pick one side of the vertex before choosing the square-root sign.

15. Circular ride diagram [7]

a. Centre height `2+3=5 m`; maximum `5+3=8 m`. [2]

b. Domain `[0,360] s`; range `[2,8] m`. [2]

c. Five metres is the centre height. Add the radius again to reach the top, giving 8 m. [1]

d. Draw one smooth cycle beginning at `(0,2)`, rising to `(180,8)`, then returning to `(360,2)`. Label time in seconds and height in metres. [2]

Thinking move: Bottom + radius = centre; centre + radius = top. Half a turn takes half the rotation time.

16. Average and instantaneous rates [8]

a. `H(0)=3`, `H(2)=-2(2)^2+16(2)+3=27`; average rate `(27-3)/2=12 m/s`. [2]

b. `H(1.9)=26.18`, `H(2.1)=27.78`; centred estimate `(27.78-26.18)/(2.1-1.9)=8 m/s`. [3]

c. Complete the square: `H(t)=-2(t-4)^2+35`, so the maximum modelled height is `35 m` at `t=4 s`. [2]

d. Part a is a secant slope across two seconds; part b estimates the tangent slope near one instant. The ball's upward rate changes with time. [1]

Thinking move: Show both heights in a centred estimate and attach m/s to each rate.

17. Graph properties and interval notation [5]

a. `r(-x)=(-x)^5-2(-x)=-x^5+2x=-(x^5-2x)=-r(x)`. Thus `r` is odd. [2]

b. Origin symmetry gives `(-2,-28)`. [1]

c. `|x-3|<=2` means `-2<=x-3<=2`, hence `1<=x<=5`, or `[1,5]`. [2]

18. Secant and tangent diagram [5]

a. Secant `AB` gives the average rate on `[3,5]`: `(5-1)/(5-3)=2` units of y per unit of `x`. [2]

b. The horizontal tangent at `A` gives instantaneous rate at `x=3`: slope `0`. [2]

c. The secant uses two points over the rising interval; the tangent gives the direction at the minimum, where the graph is flat. [1]

Thinking move: A shared point does not make the lines' slopes equal; the second point and interval matter.